

Git Version Control

Terminal-Based Assignment · Muhammad Bilal · June 2026

■ Task 1 — Initialize Repository

```
$ mkdir my-git-project
$ cd my-git-project/
$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint: git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint: git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/muhammad-bilal-tahir/my-git-project/.git/
$ ls -la
total 12
drwxrwxr-x 3 muhammad-bilal-tahir muhammad-bilal-tahir 4096 Jun 11 14:50 .
drwxr-x--- 21 muhammad-bilal-tahir muhammad-bilal-tahir 4096 Jun 11 14:50 ..
drwxrwxr-x 6 muhammad-bilal-tahir muhammad-bilal-tahir 4096 Jun 11 14:50 .git
$ ls -la .git/
total 36
drwxrwxr-x 6 muhammad-bilal-tahir muhammad-bilal-tahir 4096 Jun 11 14:50 .
drwxrwxr-x 3 muhammad-bilal-tahir muhammad-bilal-tahir 4096 Jun 11 14:50 ..
-rw-rw-r-- 1 muhammad-bilal-tahir muhammad-bilal-tahir 23 Jun 11 14:50 HEAD
-rw-rw-r-- 1 muhammad-bilal-tahir muhammad-bilal-tahir 92 Jun 11 14:50 config
-rw-rw-r-- 1 muhammad-bilal-tahir muhammad-bilal-tahir 73 Jun 11 14:50 description
drwxrwxr-x 2 muhammad-bilal-tahir muhammad-bilal-tahir 4096 Jun 11 14:50 hooks
drwxrwxr-x 2 muhammad-bilal-tahir muhammad-bilal-tahir 4096 Jun 11 14:50 info
drwxrwxr-x 4 muhammad-bilal-tahir muhammad-bilal-tahir 4096 Jun 11 14:50 objects
drwxrwxr-x 4 muhammad-bilal-tahir muhammad-bilal-tahir 4096 Jun 11 14:50 refs
$ cat .git/HEAD
ref: refs/heads/master
```

■ Task 2 — Configure Git Identity

```
$ git config --global user.name "Muhammad Bilal"
$ git config --global user.email "acc.mbilalsiddiqui@gmail.com"
$ git config --local user.name "Muhammad Bilal"
$ git config --local user.email "acc.mbilalsiddiqui@gmail.com"
```

```
$ git config --global --list
user.name=Muhammad Bilal
user.email=acc.mbilalsiddiqui@gmail.com
$ git config --local --list
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
user.name=Muhammad Bilal
user.email=acc.mbilalsiddiqui@gmail.com
$ cat ~/.gitconfig
cat: /home/muhammad-bilal-tahir/: Is a directory
cat: .gitconfig: No such file or directory
$ cat ~/.gitconfig[user]
name = Muhammad Bilal
email = acc.mbilalsiddiqui@gmail.com
$ git config user.email
acc.mbilalsiddiqui@gmail.com
```

■ Task 3 – Create & Commit Files

```
$ echo "# My Project" > README.md
```

```
$ git status
```

On branch master

No commits yet

Untracked files:

(use "git add <file>..." to include in what will be committed)

README.md

nothing added to commit but untracked files present (use "git add" to track)

```
$ git add .
```

```
$ git status
```

On branch master

No commits yet

Changes to be committed:

(use "git rm --cached <file>..." to unstage)

new file: README.md

```
$ git commit -m "Initial commit: add README"
```

[master (root-commit) ea32e32] Initial commit: add README

1 file changed, 1 insertion(+)

create mode 100644 README.md

```
$ echo "# Project Notes" > notes.txt
```

```
$ git status
```

On branch master

Untracked files:

(use "git add <file>..." to include in what will be committed)

notes.txt

nothing added to commit but untracked files present (use "git add" to track)

```
$ git add .
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   notes.txt
```

```
$ git commit -m "Initial commit: add notes.txt"
[master 494a651] Initial commit: add notes.txt
1 file changed, 1 insertion(+)
 create mode 100644 notes.txt
$ git status
On branch master
nothing to commit, working tree clean
$ echo "Version: 1.0" >> README.md
$ echo "TODO: Add tests" >> notes.txt
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   README.md
    modified:   notes.txt
```

```
no changes added to commit (use "git add" and/or "git commit -a")
$ git add README.md
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   README.md
```

```
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   notes.txt
```

```
$ git add notes.txt
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   README.md
    modified:   notes.txt
```

```
$ git commit -m "Update README and notes.txt; leave notes pending"
[master 7764fa8] Update README and notes.txt; leave notes pending
2 files changed, 2 insertions(+)
$ git status
On branch master
nothing to commit, working tree clean
```

■ Task 4 – Project Structure & .gitignore

```
$ mkdir src tests
$ echo "def greet(): return \"Hello\"" > src/greet.py
git add src/
git commit -m "feat: add greeting function in src/greet.py"

[master eb283a1] feat: add greeting function in src/greet.py
1 file changed, 1 insertion(+)
create mode 100644 src/greet.py

$ echo "[settings]\ndebug=False" > config.ini
git add config.ini
git commit -m "config: add application settings file"
[master 902cdc7] config: add application settings file
1 file changed, 1 insertion(+)
create mode 100644 config.ini

$ echo "import greet\nassert greet.greet() == \"Hello\"" > tests/test_greet.py
git add tests/
git commit -m "test: add unit test for greet function"
[master 70580ac] test: add unit test for greet function
1 file changed, 1 insertion(+)
create mode 100644 tests/test_greet.py

$
echo "## Installation\npip install -r requirements.txt" >> README.md
git add README.md
git commit -m "docs: update README with installation instructions"
[master c66b32b] docs: update README with installation instructions
1 file changed, 1 insertion(+)

$ echo "def greet(name=\"World\"): return f\"Hello, {name}\"" > src/greet.py
git add src/greet.py
git commit -m "fix: greet() now accepts optional name parameter"
[master 46a5ef1] fix: greet() now accepts optional name parameter
1 file changed, 1 insertion(+), 1 deletion(-)

$ git log --oneline
46a5ef1 (HEAD -> master) fix: greet() now accepts optional name parameter
c66b32b docs: update README with installation instructions
70580ac test: add unit test for greet function
902cdc7 config: add application settings file
eb283a1 feat: add greeting function in src/greet.py
7764fa8 Update README and notes.txt; leave notes pending
494a651 Initial commit: add notes.txt
ea32e32 Initial commit: add README

$ git log --oneline
46a5ef1 (HEAD -> master) fix: greet() now accepts optional name parameter
c66b32b docs: update README with installation instructions
70580ac test: add unit test for greet function
902cdc7 config: add application settings file
eb283a1 feat: add greeting function in src/greet.py
7764fa8 Update README and notes.txt; leave notes pending
```

```
494a651 Initial commit: add notes.txt
ea32e32 Initial commit: add README
$ git log -3 --stat
commit 46a5ef1302b9bb75253548e381ab6f6119b5fbc (HEAD -> master)
Author: Muhammad Bilal <acc.mbilalsiddiqui@gmail.com>
Date: Thu Jun 11 15:35:08 2026 +0500
```

```
fix: greet() now accepts optional name parameter
```

```
src/greet.py | 2 +-
1 file changed, 1 insertion(+), 1 deletion(-)
```

```
commit c66b32b67ed1cf85ecd96e1b3e51fbf8660f5458
Author: Muhammad Bilal <acc.mbilalsiddiqui@gmail.com>
Date: Thu Jun 11 15:34:45 2026 +0500
```

```
docs: update README with installation instructions
```

```
README.md | 1 +
1 file changed, 1 insertion(+)
```

```
commit 70580ac0e1c082e3b45d071789402a23cab83e88
Author: Muhammad Bilal <acc.mbilalsiddiqui@gmail.com>
Date: Thu Jun 11 15:34:11 2026 +0500
```

```
test: add unit test for greet function
```

```
tests/test_greet.py | 1 +
1 file changed, 1 insertion(+)
```

```
$ git show HEAD
commit 46a5ef1302b9bb75253548e381ab6f6119b5fbc (HEAD -> master)
Author: Muhammad Bilal <acc.mbilalsiddiqui@gmail.com>
Date: Thu Jun 11 15:35:08 2026 +0500
```

```
fix: greet() now accepts optional name parameter
```

```
diff --git a/src/greet.py b/src/greet.py
index c2e3f33..a36698a 100644
--- a/src/greet.py
+++ b/src/greet.py
@@ -1,1 +1 @@
-def greet(): return "Hello"
+def greet(name="World"): return f"Hello, {name}"
$ git log -1 --format='%H'
46a5ef1302b9bb75253548e381ab6f6119b5fbc
$ it log --oneline --graph --all
it: command not found
```

■ Task 5 – View Commit History

```
$ git log --oneline --graph --all
* 46a5ef1 (HEAD -> master) fix: greet() now accepts optional name parameter
* c66b32b docs: update README with installation instructions
* 70580ac test: add unit test for greet function
* 902cdc7 config: add application settings file
* eb283a1 feat: add greeting function in src/greet.py
```

```
* 7764fa8 Update README and notes.txt; leave notes pending
* 494a651 Initial commit: add notes.txt
* ea32e32 Initial commit: add README
$ echo "print(\"New feature added\")" >> notes.txt
echo "## Changelog\n- v1.1 released" >> README.md
$ git diff
diff --git a/README.md b/README.md
index 1323ccf..9a191fe 100644
--- a/README.md
+++ b/README.md
@@ -1,3 +1,4 @@
# My Project
Version: 1.0
## Installation\npip install -r requirements.txt
+## Changelog\n- v1.1 released
diff --git a/notes.txt b/notes.txt
index f4dad48..0fcc36e 100644
--- a/notes.txt
+++ b/notes.txt
@@ -1,2 +1,3 @@
# Project Notes
TODO: Add tests
+print("New feature added")
$ git add README.md
$ git diff
diff --git a/notes.txt b/notes.txt
index f4dad48..0fcc36e 100644
--- a/notes.txt
+++ b/notes.txt
@@ -1,2 +1,3 @@
# Project Notes
TODO: Add tests
+print("New feature added")
$ git diff --stage
error: invalid option: --stage
usage: git diff [<options>] [<commit>] [--] [<path>...]
or: git diff [<options>] --cached [--merge-base] [<commit>] [--] [<path>...]
or: git diff [<options>] [--merge-base] <commit> [<commit>...] <commit> [--] [<path>...]
or: git diff [<options>] <commit>...<commit> [--] [<path>...]
or: git diff [<options>] <blob> <blob>
or: git diff [<options>] --no-index [--] <path> <path> [<paths-spec>...]

common diff options:
-z output diff-raw with lines terminated with NUL.
-p output patch format.
-u synonym for -p.
--patch-with-raw
output both a patch and the diff-raw format.
```

```

--stat show diffstat instead of patch.
--numstat show numeric diffstat instead of patch.
--patch-with-stat
output a patch and prepend its diffstat.
--name-only show only names of changed files.
--name-status show names and status of changed files.
--full-index show full object name on index lines.
--abbrev=<n> abbreviate object names in diff-tree header and diff-raw.
-R swap input file pairs.
-B detect complete rewrites.
-M detect renames.
-C detect copies.
--find-copies-harder
try unchanged files as candidate for copy detection.
-l<n> limit rename attempts up to <n> paths.
-O<file> reorder diffs according to the <file>.
-S<string> find filepair whose only one side contains the string.
--pickaxe-all
show all files diff when -S is used and hit is found.
-a --text treat all files as text.

```

```
$ git diff --staged
```

```

diff --git a/README.md b/README.md
index 1323ccf..9a191fe 100644
--- a/README.md
+++ b/README.md
@@ -1,3 +1,4 @@
# My Project
Version: 1.0
## Installation\npip install -r requirements.txt
+## Changelog\n- v1.1 released
$ echo "def add(a, b): return a + b" >> src/greet.py
echo "assert add(2,3) == 5" >> tests/test_greet.py
echo "DEBUG=True # dev mode" >> config.ini

```

```
$ git status
```

```

On branch master
Changes to be committed:
(use "git restore --staged <file>..." to unstage)
modified: README.md

```

```

Changes not staged for commit:
(use "git add <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
modified: config.ini
modified: notes.txt
modified: src/greet.py
modified: tests/test_greet.py

```

```
$ git diff
```

```
diff --git a/config.ini b/config.ini
```

```
index 9b61cdf..4890693 100644
--- a/config.ini
+++ b/config.ini
@@ -1,2 @@
[settings]\nDEBUG=False
+DEBUG=True # dev mode
diff --git a/notes.txt b/notes.txt
index f4dad48..0fcc36e 100644
--- a/notes.txt
+++ b/notes.txt
@@ -1,2 +1,3 @@
# Project Notes
TODO: Add tests
+print("New feature added")
diff --git a/src/greet.py b/src/greet.py
index a36698a..a326989 100644
--- a/src/greet.py
+++ b/src/greet.py
@@ -1 +1,2 @@
def greet(name="World"): return f"Hello, {name}"
+def add(a, b): return a + b
diff --git a/tests/test_greet.py b/tests/test_greet.py
index 8d455be..c6175ad 100644
--- a/tests/test_greet.py
+++ b/tests/test_greet.py
@@ -1 +1,2 @@
import greet\nassert greet.greet() == "Hello"
+assert add(2,3) == 5
$ git add -A
$ git diff --staged
diff --git a/README.md b/README.md
index 1323ccf..9a191fe 100644
--- a/README.md
+++ b/README.md
@@ -1,3 +1,4 @@
# My Project
Version: 1.0
## Installation\npip install -r requirements.txt
+## Changelog\n- v1.1 released
diff --git a/config.ini b/config.ini
index 9b61cdf..4890693 100644
--- a/config.ini
+++ b/config.ini
@@ -1 +1,2 @@
[settings]\nDEBUG=False
+DEBUG=True # dev mode
diff --git a/notes.txt b/notes.txt
index f4dad48..0fcc36e 100644
```



```
--- a/notes.txt
+++ b/notes.txt
@@ -1,2 +1,3 @@
# Project Notes
TODO: Add tests
+print("New feature added")
diff --git a/src/greet.py b/src/greet.py
index a36698a..a326989 100644
--- a/src/greet.py
+++ b/src/greet.py
@@ -1 +1,2 @@
def greet(name="World"): return f"Hello, {name}"
+def add(a, b): return a + b
diff --git a/tests/test_greet.py b/tests/test_greet.py
index 8d455be..c6175ad 100644
--- a/tests/test_greet.py
+++ b/tests/test_greet.py
@@ -1 +1,2 @@
import greet\nassert greet.greet() == "Hello"
+assert add(2,3) == 5
$ git commit -m "feat: add math utilities and update tests and config"
[master e5d576e] feat: add math utilities and update tests and config
5 files changed, 5 insertions(+)
$ mkdir -p src tests docs
echo "# Calculator App" > README.md
echo "def add(a,b): return a+b" > src/calc.py
git add .
git commit -m "init: project scaffold with calculator module"
[master 8b50362] init: project scaffold with calculator module
2 files changed, 2 insertions(+), 4 deletions(-)
create mode 100644 src/calc.py
$ echo "def sub(a,b): return a-b" >> src/calc.py
echo "def mul(a,b): return a*b" >> src/calc.py
git add src/calc.py
git commit -m "feat: add subtraction and multiplication functions"
[master c718cbb] feat: add subtraction and multiplication functions
1 file changed, 2 insertions(+)
$ echo "from src.calc import add,sub,mul" > tests/test_calc.py
echo "assert add(1,2)==3\nassert sub(5,3)==2" >> tests/test_calc.py
git add tests/
git commit -m "test: add unit tests for add and sub functions"
[master ded9a8f] test: add unit tests for add and sub functions
1 file changed, 2 insertions(+)
create mode 100644 tests/test_calc.py
$ cho "def div(a,b):\n if b==0: raise ValueError(\"Cannot divide by zero\")\n return a/b" >> src/calc.py
git add src/calc.py
git commit -m "feat: add division with zero-division error handling"echo "from src.calc import add,sub,mul" >
tests/test_calc.py
```

```

echo "assert add(1,2)==3\nassert sub(5,3)==2" >> tests/test_calc.py
Command 'cho' not found, did you mean:
command 'cht' from deb chemtool (1.6.14-6build3)
command 'chg' from deb mercurial (7.2-3build1)
command 'co' from deb rcs (5.10.1-3)
command 'who' from deb coreutils-from-gnu (0.0.0-ubuntu25)
command 'who' from deb coreutils-from-uutils (0.0.0-ubuntu25)
command 'who' from deb coreutils-from-busybox (0.0.0-ubuntu25)
command 'who' from deb coreutils-from-toybox (0.0.0-ubuntu25)
command 'cdo' from deb cdo (2.5.4-1)
command 'chr' from deb chr (0.1.80-1)
command 'chr' from deb chr-tiny (0.1.80-1)
command 'echo' from deb coreutils-from-gnu (0.0.0-ubuntu25)
command 'echo' from deb coreutils-from-uutils (0.0.0-ubuntu25)
command 'echo' from deb coreutils-from-busybox (0.0.0-ubuntu25)
command 'echo' from deb coreutils-from-toybox (0.0.0-ubuntu25)
Try: sudo apt install <deb name>
error: pathspec 'from src.calc import add,sub,mul' did not match any file(s) known to git
$ git add tests/
git commit -m "test: add unit tests for add and sub functions"
[master b0d0875] test: add unit tests for add and sub functions
1 file changed, 1 deletion(-)
$ echo "assert add(1,2)==3\nassert sub(5,3)==2" >> tests/test_calc.py
$ git commit -m "test: add unit tests for add and sub functions"
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
modified: tests/test_calc.py

```

```

no changes added to commit (use "git add" and/or "git commit -a")
$
git log --oneline --graph
* b0d0875 (HEAD -> master) test: add unit tests for add and sub functions
* ded9a8f test: add unit tests for add and sub functions
* c718cbb feat: add subtraction and multiplication functions
* 8b50362 init: project scaffold with calculator module
* e5d576e feat: add math utilities and update tests and config
* 46a5ef1 fix: greet() now accepts optional name parameter
* c66b32b docs: update README with installation instructions
* 70580ac test: add unit test for greet function
* 902cdc7 config: add application settings file
* eb283a1 feat: add greeting function in src/greet.py
* 7764fa8 Update README and notes.txt; leave notes pending
* 494a651 Initial commit: add notes.txt
* ea32e32 Initial commit: add README
$ git log --stat --oneline

```

```

b0d0875 (HEAD -> master) test: add unit tests for add and sub functions

```

```
tests/test_calc.py | 1 -
1 file changed, 1 deletion(-)
ded9a8f test: add unit tests for add and sub functions
tests/test_calc.py | 2 ++
1 file changed, 2 insertions(+)
c718cbb feat: add subtraction and multiplication functions
src/calc.py | 2 ++
1 file changed, 2 insertions(+)
8b50362 init: project scaffold with calculator module
README.md | 5 +----
src/calc.py | 1 +
2 files changed, 2 insertions(+), 4 deletions(-)
e5d576e feat: add math utilities and update tests and config
README.md | 1 +
config.ini | 1 +
notes.txt | 1 +
src/greet.py | 1 +
tests/test_greet.py | 1 +
5 files changed, 5 insertions(+)
46a5ef1 fix: greet() now accepts optional name parameter
src/greet.py | 2 +-
1 file changed, 1 insertion(+), 1 deletion(-)
c66b32b docs: update README with installation instructions
README.md | 1 +
1 file changed, 1 insertion(+)
70580ac test: add unit test for greet function
tests/test_greet.py | 1 +
1 file changed, 1 insertion(+)
902cdc7 config: add application settings file
config.ini | 1 +
1 file changed, 1 insertion(+)
eb283a1 feat: add greeting function in src/greet.py
src/greet.py | 1 +
1 file changed, 1 insertion(+)
7764fa8 Update README and notes.txt; leave notes pending
README.md | 1 +
notes.txt | 1 +
2 files changed, 2 insertions(+)
494a651 Initial commit: add notes.txt
notes.txt | 1 +
1 file changed, 1 insertion(+)
ea32e32 Initial commit: add README
README.md | 1 +
1 file changed, 1 insertion(+)
$ git log --oneline
b0d0875 (HEAD -> master) test: add unit tests for add and sub functions
ded9a8f test: add unit tests for add and sub functions
c718cbb feat: add subtraction and multiplication functions
```

```

8b50362 init: project scaffold with calculator module
e5d576e feat: add math utilities and update tests and config
46a5ef1 fix: greet() now accepts optional name parameter
c66b32b docs: update README with installation instructions
70580ac test: add unit test for greet function
902cdc7 config: add application settings file
eb283a1 feat: add greeting function in src/greet.py
7764fa8 Update README and notes.txt; leave notes pending
494a651 Initial commit: add notes.txt
ea32e32 Initial commit: add README
$ git diff z1a2b3c b3c4d5e
fatal: ambiguous argument 'z1a2b3c': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'

```

■ Task 6 – File Recovery via Checkout

```

$ echo "" > src/calc.py
cat src/calc.py

```

```

$ git log --oneline src/calc.py
c718cbb feat: add subtraction and multiplication functions
8b50362 init: project scaffold with calculator module
$ git checkout a2b3c4d -- src/calc.py
fatal: invalid reference: a2b3c4d
$ git log --oneline
b0d0875 (HEAD -> master) test: add unit tests for add and sub functions
ded9a8f test: add unit tests for add and sub functions
c718cbb feat: add subtraction and multiplication functions
8b50362 init: project scaffold with calculator module
e5d576e feat: add math utilities and update tests and config
46a5ef1 fix: greet() now accepts optional name parameter
c66b32b docs: update README with installation instructions
70580ac test: add unit test for greet function
902cdc7 config: add application settings file
eb283a1 feat: add greeting function in src/greet.py
7764fa8 Update README and notes.txt; leave notes pending
494a651 Initial commit: add notes.txt
ea32e32 Initial commit: add README
$ cd ..

```

■ Task 7 – Team Project & Remote Setup

```

$ mkdir team-project && cd team-project
$ git init
echo "# Team Project" > README.md
git add .
git commit -m "init: project start"
git log --oneline
hint: Using 'master' as the name for the initial branch. This default branch name

```

```
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint: git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint: git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/muhammad-bilal-tahir/team-project/.git/
[master (root-commit) 9093a20] init: project start
1 file changed, 1 insertion(+)
create mode 100644 README.md
9093a20 (HEAD -> master) init: project start
$
git remote add origin https://github.com/muhammad-bilal-tahir/Neon.git
git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/muhammad-bilal-tahir/Neon.git'
$ git push -u origin master
Username for 'https://github.com':
Password for 'https://github.com':
remote: No anonymous write access.
fatal: Authentication failed for 'https://github.com/muhammad-bilal-tahir/Neon.git/'
$ git clone https://github.com/teamuser/team-project.git
cd team-project
git log --onelinegit init
Cloning into 'team-project'...
Username for 'https://github.com':
Password for 'https://github.com':
remote: Repository not found.
fatal: Authentication failed for 'https://github.com/teamuser/team-project.git/'
bash: cd: team-project: No such file or directory
fatal: ambiguous argument 'init': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'
$ git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/muhammad-bilal-tahir/Neon.git
git push -u origin master
On branch master
nothing to commit, working tree clean
error: remote origin already exists.
error: src refspec master does not match any
```

```
error: failed to push some refs to 'https://github.com/muhammad-bilal-tahir/Neon.git'
$ git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/muhammad-bilal-tahir/Neon.git
git push -u origin main
On branch main
nothing to commit, working tree clean
error: remote origin already exists.
Username for 'https://github.com':
Password for 'https://github.com':
remote: No anonymous write access.
fatal: Authentication failed for 'https://github.com/muhammad-bilal-tahir/Neon.git/'
$ git push -u origin main
Username for 'https://github.com': muhammad-bilal-tahir
Password for 'https://muhammad-bilal-tahir@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/muhammad-bilal-tahir/Neon.git/'
$ git push -u origin main
Username for 'https://github.com': muhammad-bilal-tahir
Password for 'https://muhammad-bilal-tahir@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 242 bytes | 242.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/muhammad-bilal-tahir/Neon.git
* [new branch] main -> main
branch 'main' set up to track 'origin/main'.
$ git clone https://github.com/muhammad-bilal-tahir/Neon.git
cd team-project
git log --onelinegit init
Cloning into 'Neon'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
bash: cd: team-project: No such file or directory
fatal: ambiguous argument 'init': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'
$ git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/muhammad-bilal-tahir/Neon.git
git push -u origin main
On branch main
Your branch is up to date with 'origin/main'.
```

Untracked files:

(use "git add <file>..." to include in what will be committed)

Neon/

nothing added to commit but untracked files present (use "git add" to track)

error: remote origin already exists.

Username for 'https://github.com': muhammad-bilal-tahir

Password for 'https://muhammad-bilal-tahir@github.com':

branch 'main' set up to track 'origin/main'.

Everything up-to-date

\$ cd ..

\$ mkdir local-project && cd local-project

git init

echo "Hello Git" > hello.txt

git add .

git commit -m "init: first commit"

hint: Using 'master' as the name for the initial branch. This default branch name

hint: will change to "main" in Git 3.0. To configure the initial branch name

hint: to use in all of your new repositories, which will suppress this warning,

hint: call:

hint:

hint: git config --global init.defaultBranch <name>

hint:

hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and

hint: 'development'. The just-created branch can be renamed via this command:

hint:

hint: git branch -m <name>

hint:

hint: Disable this message with "git config set advice.defaultBranchName false"

Initialized empty Git repository in /home/muhammad-bilal-tahir/local-project/.git/

[master (root-commit) 417b5c6] init: first commit

1 file changed, 1 insertion(+)

create mode 100644 hello.txt

\$ git remote add origin https://github.com/muhammad-bilal-tahir/Neon.git

\$ git remote -v

origin https://github.com/muhammad-bilal-tahir/Neon.git (fetch)

origin https://github.com/muhammad-bilal-tahir/Neon.git (push)

\$ git remote show origin

* remote origin

Fetch URL: https://github.com/muhammad-bilal-tahir/Neon.git

Push URL: https://github.com/muhammad-bilal-tahir/Neon.git

HEAD branch: main

Remote branch:

main new (next fetch will store in remotes/origin)

\$ git remote rename origin upstream

\$ git remote remove upstream

\$ echo "v2.0 changes" >> README.md

git add README.md

git commit -m "docs: update README for v2.0"

```
[master e5d0150] docs: update README for v2.0
1 file changed, 1 insertion(+)
create mode 100644 README.md

$ git status

On branch master

nothing to commit, working tree clean

$ cd ..

$ cd team-project/

$ echo "v2.0 changes" >> README.md

git add README.md

git commit -m "docs: update README for v2.0"

[main e1b119a] docs: update README for v2.0
1 file changed, 1 insertion(+)

$ git status

On branch main

Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)
```

```
Untracked files:
(use "git add <file>..." to include in what will be committed)

Neon/
```

nothing added to commit but untracked files present (use "git add" to track)

```
$ git push origin main

Username for 'https://github.com': muhammad-bilal-tahir
Password for 'https://muhammad-bilal-tahir@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 298 bytes | 298.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/muhammad-bilal-tahir/Neon.git
9093a20..e1b119a main -> main

$ git status

On branch main

Your branch is up to date with 'origin/main'.
```

```
Untracked files:
(use "git add <file>..." to include in what will be committed)

Neon/
```

nothing added to commit but untracked files present (use "git add" to track)

```
$ git log --onilne origin/main
fatal: unrecognized argument: --onilne

$ git log --onilne origin/master
fatal: ambiguous argument 'origin/master': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'

$ git log --online origin/main

e1b119a (HEAD -> main, origin/main) docs: update README for v2.0
9093a20 init: project start
```


■ Task 8 – Feature Branch & Push

```
$ git checkout -b feature/login
echo "login logic" > login.py
git add login.py
git commit -m "feat: add login module"
git push -u origin feature/login
Switched to a new branch 'feature/login'
[feature/login cb35c02] feat: add login module
1 file changed, 1 insertion(+)
create mode 100644 login.py
Username for 'https://github.com': muhammad-bilal-tahir
Password for 'https://muhammad-bilal-tahir@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 302 bytes | 302.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/login' on GitHub by visiting:
remote: https://github.com/muhammad-bilal-tahir/Neon/pull/new/feature/login
remote:
To https://github.com/muhammad-bilal-tahir/Neon.git
* [new branch] feature/login -> feature/login
branch 'feature/login' set up to track 'origin/feature/login'.
$
cat src/calc.py
git log --oneline src/calc.py
cat: src/calc.py: No such file or directory
fatal: ambiguous argument 'src/calc.py': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'
$ cd ..
$
cat src/calc.py
git log --oneline src/calc.py
cat: src/calc.py: No such file or directory
fatal: not a git repository (or any of the parent directories): .git
$ cd team-project/
$ git log --oneline
cb35c02 (HEAD -> feature/login, origin/feature/login) feat: add login module
e1b119a (origin/main, main) docs: update README for v2.0
9093a20 init: project start
```

■ Task 9 – Refs, Reflog & Tags

```
$ git show-ref
cb35c02f38b3410306d774bd1e8a9a8274c037b9 refs/heads/feature/login
```

```
e1b119ab95a9e4a5381bce6f9e32591decbb6934 refs/heads/main
cb35c02f38b3410306d774bd1e8a9a8274c037b9 refs/remotes/origin/feature/login
e1b119ab95a9e4a5381bce6f9e32591decbb6934 refs/remotes/origin/main
$ cat .git/HEAD
ref: refs/heads/feature/login
$ git reflog --all
cb35c02 (HEAD -> feature/login, origin/feature/login) refs/remotes/origin/feature/login@{0}: update by push
cb35c02 (HEAD -> feature/login, origin/feature/login) refs/heads/feature/login@{0}: commit: feat: add login module
e1b119a (origin/main, main) refs/heads/feature/login@{1}: branch: Created from HEAD
cb35c02 (HEAD -> feature/login, origin/feature/login) HEAD@{0}: commit: feat: add login module
e1b119a (origin/main, main) HEAD@{1}: checkout: moving from main to feature/login
e1b119a (origin/main, main) refs/remotes/origin/main@{0}: update by push
e1b119a (origin/main, main) refs/heads/main@{0}: commit: docs: update README for v2.0
e1b119a (origin/main, main) HEAD@{2}: commit: docs: update README for v2.0
9093a20 refs/heads/main@{1}: Branch: renamed refs/heads/main to refs/heads/main
9093a20 HEAD@{3}: Branch: renamed refs/heads/main to refs/heads/main
9093a20 refs/remotes/origin/main@{1}: update by push
9093a20 refs/heads/main@{2}: Branch: renamed refs/heads/main to refs/heads/main
9093a20 HEAD@{5}: Branch: renamed refs/heads/main to refs/heads/main
9093a20 refs/heads/main@{3}: Branch: renamed refs/heads/master to refs/heads/main
9093a20 HEAD@{7}: Branch: renamed refs/heads/master to refs/heads/main
9093a20 refs/heads/main@{4}: commit (initial): init: project start
9093a20 HEAD@{9}: commit (initial): init: project start
$ git show HEAD@{2}
commit e1b119ab95a9e4a5381bce6f9e32591decbb6934 (origin/main, main)
Author: Muhammad Bilal <acc.mbilalsiddiqui@gmail.com>
Date: Thu Jun 11 17:08:35 2026 +0500
```

```
docs: update README for v2.0
```

```
diff --git a/README.md b/README.md
index 7239262..20afba1 100644
--- a/README.md
+++ b/README.md
@@ -1,2 @@
# Team Project
+v2.0 changes
```

```
$ git show HEAD~1
git show HEAD~2
commit e1b119ab95a9e4a5381bce6f9e32591decbb6934 (origin/main, main)
Author: Muhammad Bilal <acc.mbilalsiddiqui@gmail.com>
Date: Thu Jun 11 17:08:35 2026 +0500
```

```
docs: update README for v2.0
```

```
diff --git a/README.md b/README.md
index 7239262..20afba1 100644
--- a/README.md
+++ b/README.md
@@ -1,2 @@
```

```
# Team Project
+v2.0 changes
commit 9093a20625a011f1557c95190fcad8296283e470
Author: Muhammad Bilal <acc.mbilalsiddiqui@gmail.com>
Date: Thu Jun 11 16:35:33 2026 +0500
```

```
init: project start
```

```
diff --git a/README.md b/README.md
new file mode 100644
index 0000000..7239262
--- /dev/null
+++ b/README.md
@@ -0,0 +1 @@
+# Team Project

$ git tag -a v1.0 -m "Version 1.0 release"
git tag
git show v1.0git show HEAD~1
git show HEAD~2
v1.0
fatal: ambiguous argument 'v1.0git': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'
commit 9093a20625a011f1557c95190fcad8296283e470
Author: Muhammad Bilal <acc.mbilalsiddiqui@gmail.com>
Date: Thu Jun 11 16:35:33 2026 +0500
```

```
init: project start
```

```
diff --git a/README.md b/README.md
new file mode 100644
index 0000000..7239262
--- /dev/null
+++ b/README.md
@@ -0,0 +1 @@
+# Team Project

$ git branch -v
* feature/login cb35c02 feat: add login module
main e1b119a docs: update README for v2.0

$ echo "CRITICAL_FEATURE=True" >> config.ini
git add config.inigit branch -v
fatal: pathspec 'config.inigit' did not match any files

$ git fsck --lost-found
Checking ref database: 100% (1/1), done.
Checking object directories: 100% (256/256), done.

$ echo "Important work" >> notes.txt
git stash push -m 'Important feature work'
No local changes to save

$ git stash list
git stash pop
echo "Important work" >> app.py
```

```
git stash push -m 'Important feature work'
No stash entries found.
No local changes to save
$ git stash list
git stash pop
echo "Important work" >> notes.txt
git stash push -m 'Important feature work'
No stash entries found.
No local changes to save
```

■ Task 10 – Branch Delete & Recovery

```
$ git checkout -b feature/payment
echo "def pay(): pass" > payment.py
git add payment.py
git commit -m "feat: add payment module"
echo "def refund(): pass" >> payment.py
git add payment.py
git commit -m "feat: add refund function"
```

```
Switched to a new branch 'feature/payment'
[feature/payment f61cf70] feat: add payment module
1 file changed, 1 insertion(+)
create mode 100644 payment.py
[feature/payment 4550a60] feat: add refund function
1 file changed, 1 insertion(+)
```

```
$ git checkout main
git branch -D feature/payment
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Deleted branch feature/payment (was 4550a60).
$ git reflog | grep paymentgit checkout main
git branch -D feature/payment
grep: checkout: No such file or directory
grep: main: No such file or directory
error: branch 'feature/payment' not found
$ git reflog | grep paymentgit checkout master
git branch -D feature/payment
```

```
grep: checkout: No such file or directory
grep: master: No such file or directory
error: branch 'feature/payment' not found
$ git log --oneline
e1b119a (HEAD -> main, origin/main) docs: update README for v2.0
9093a20 init: project start
$ cd ..
$
```